

MATH 110

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1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t)$, $t \in \mathbb{R}$

$$\frac{du}{dt} = u(1 - u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{du}{dt} = \mathbf{u} \text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned} \text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ &\text{etc.} \end{aligned}$$

Examples of PDEs

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

ex Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x , time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l, k$ constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x, 0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2nd order
- A general 2nd order PDE for $u(x, t)$ is:

$$G(x, t, u, u_x, u_t, u_{xx}, u_{tt}, u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u . It is nonhomogeneous/inhomogeneous otherwise.

ex Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2t}u_{xt} = 0$
is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + \sin(u) = 0$
is nonlinear because of the $\sin(u)$ term.

Operating notation & linearity

We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x, t)$ to produce a new function.

$$\begin{aligned} L(u) &= Lu = (\partial_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u + v) &= Lu + Lv \\ \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ \forall \text{ constants } c \in \mathbb{R} \\ \forall \text{ functions } u \end{aligned}$$

[ex] Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u + v) &= (\partial_t + k\partial_x^2)(u + v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is

- homogeneous if $f = 0$
- inhomogeneous if $f \neq 0$

- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b .

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

Solutions to PDEs

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

[ex] Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x, t) &= x^2 + 2t \text{ is a soln} \\ \rightarrow \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x, t) &= e^{-t} \sin x \text{ is a soln} \\ \rightarrow \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t} \sin x + e^{-t} \sin x = 0 \end{aligned}$$

[ex] Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu &= f \text{ with } L = \partial_x \\ f(x, t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x .

$$u(x, t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

[ex] Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \rightarrow L &= (\partial_t^2 - 4) \\ f(x, t) &= 0 \end{aligned}$$

There is no explicit x -dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x, t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 - Characteristics

Recap:

- Heat equation for $u(x, t)$

$$u_t - ku_{xx} = 0, \quad k > 0, \text{ constant}$$

- We can write the heat equation as $Lu = 0$, where $L = (\partial_t - k\partial_x^2)$ is a differential operator.
- L is linear $\leftrightarrow L(u + v) = Lu + Lv$ and $L(cu) = cLu \quad \forall$ functions u, v , constant c
- PDE: $Lu = f$ is linear if L is linear
- Linear PDE: $Lu = f$ is homogeneous if $f = 0$, inhomogeneous otherwise.

First-order linear PDE

Consider functions $u(x, y)$

$$u : \mathbb{R}^2 \rightarrow \mathbb{R}$$

Often we think of x, y as spatial variable and t as time variables. The simplest PDE we can have following this is:

$$u_x = 0$$

Which is 1st-order, linear, homogeneous. Also note that there is no explicit y -dependence. Therefore we can treat this like an ODE, and solve with direct integration of both sides.

$$u(x, y) = f(y)$$

where $f(y)$ is an arbitrary function of y .

Now consider $au_x + bu_y = 0$, which is a 1st-order, linear, homogeneous PDE with constants $a, b \in \mathbb{R}$.

$$\begin{aligned} \text{Let } V &= (a, b) \in \mathbb{R}^2 \\ \text{Recall } \nabla u &= (u_x, u_y) \\ \text{Then } V \cdot \nabla u &= (a, b) \cdot (u_x, u_y) \\ &= au_x + bu_y = 0 \end{aligned}$$

$\Rightarrow$ Directional derivative in direction V is 0.

$\Rightarrow u$ is constant on lines parallel to V .

Let $V^\perp = (b, -a)$, so that $V \cdot V^\perp = 0$.

Lines parallel to V satisfy

$$\begin{aligned} V^\perp \cdot (x, y) &= \text{constant} \\ \Rightarrow bx - ay &= \text{constant} \end{aligned}$$

Then u is constant on lines parallel to V .

$$\begin{aligned} \Rightarrow u(x, y) &= f(bx - ay) \\ \text{for an arbitrary function of } f \end{aligned}$$

Check:

$$\begin{aligned} (a\partial_x + b\partial_y) f(bx - ay) \\ = af' - bf' = 0 \\ \Rightarrow f' \text{ is a derivative of } f : \mathbb{R} \rightarrow \mathbb{R} \end{aligned}$$

Characteristic Coordinates. In this example, the lines parallel to V are called characteristics. Here u is constant along the characteristics.

We can write points (x, y) in characteristic coordinates (ξ, η) . Where ξ is the x -intercept of the characteristic line, and η is the distance along that line

to the point (x, y) .

$$\text{So } x = \xi + a\eta = x(\xi, \eta)$$

$$y = b\eta = y(\xi, \eta)$$

$$\xi = x - a\eta$$

$$\eta = \frac{y}{b}$$

$$\xi = x - \frac{a}{b}y$$

$$b\xi = bx - ay$$

Transform PDE to (ξ, η) coordinates.

Chain rule:

$$\begin{aligned} \frac{\partial u}{\partial \eta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \eta} \\ \Rightarrow u_\eta &= au_x + bu_y = 0 \end{aligned}$$

Using characteristic coordinates (ξ, η) , we've transformed to the PDE $u_\eta = 0$ for $u(\xi, \eta)$.

Solution:

$$u = f(\xi)$$

$$\text{where } \xi = \frac{1}{b}(bx - ay)$$

$$\Rightarrow u = g(bx - ay)$$

Where g is an arbitrary function.

Remarks:

- η is coordinates along characteristics
- u is constant on characteristics $\Rightarrow u_\eta = 0$
- ξ labels different characteristics. There are other choices for ξ but any choice $\tilde{\xi}$ would have $\tilde{\xi} = h(\xi)$

Method of Characteristics

Method for solving PDEs of form

$$a(x, y)u_x + b(x, y)u_y = 0 \quad \text{for } u(x, y)$$

Idea: Let $A(x, y) = (a(x, y), b(x, y))$

The directional derivative $A \cdot \nabla u = 0$

$\Rightarrow u$ is constant along curves w/ slope A .

- These are characteristic curves.
- Find coord η along chars
- Find coord ξ which labels chars
- PDE transforms to $u_\eta = 0$, general soln: $u = f(\xi)$

ex $u_x + yu_y = 0$

The directional derivative along $(1, y)$ are 0.

The chars $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = 1, \quad \frac{dy}{d\eta} = y$$

At $\eta = 0$, take $(x(0), y(0)) = (0, \xi)$

Solve:

$$\frac{dx}{d\eta} = 1, \quad x(0) = 0$$

$$\Rightarrow x = \eta$$

$$\frac{dy}{d\eta} = y, \quad y(0) = \xi$$

$$\Rightarrow y = \xi e^\eta$$

This follows that $\xi = ye^{-x}$.

$$\Rightarrow u_\eta = u_x \frac{\partial x}{\partial \eta} + u_y \frac{\partial y}{\partial \eta}$$

$$= u_x + \xi e^\eta u_y$$

$$= u_x + yu_y = 0$$

$$\Rightarrow u_\eta = 0$$

$$\Rightarrow u = f(\xi) = f(ye^{-x}) \text{ with arbitrary function } f$$

General solution:

Suppose we are given $u(0, y) = y^2$

$\Rightarrow$ Specific solution

$$u(0, y) = f(y) = y^2$$

$$\Rightarrow u(x, y) = (ye^{-x})^2$$

Alternatively, from parametric equations,

$$\frac{dx}{d\eta} = 1 \quad \frac{dy}{d\eta} = y$$

$$\Rightarrow \frac{dy}{dx} = y$$

$$\Rightarrow y = Ce^x$$

$\Rightarrow u$ is constant along chars.

$$\Rightarrow u(x, Ce^x) = u(0, C)$$

$$= u(0, ye^{-x})$$

$$= f(e^{-x}y) \text{ for arbitrary } f$$

$$\boxed{\text{ex}} \quad u_x + 2xy^2u_y = 0$$

Characteristic curves: $(x(\eta), y(\eta))$

$$\frac{dx}{d\eta} = 1 \quad \frac{dy}{d\eta} = 2xy^2$$

$$\frac{dy}{dx} = 2xy^2$$

$$\int \frac{1}{y^2} dy = \int 2x dx$$

$$-\frac{1}{y} = x^2 - c$$

$$y = \frac{1}{c - x^2}$$

u is constant along chars.

$\Rightarrow u(x, y) = f(c) = f(x^2 + \frac{1}{y})$ is the general soln.

Advection, transport

Consider $u(x, t)$ describing quantity u at position x , time t .

$$u_t + cu_x = 0 \quad (c > 0)$$

Chars: $(t(\eta), x(\eta))$ satisfy

$$\begin{aligned} \frac{dt}{d\eta} &= 1 & \frac{dx}{d\eta} &= c \\ \Rightarrow \frac{dx}{dt} &= c \\ \Rightarrow x &= ct + \xi \text{ via integration} \end{aligned}$$

General solution:

$$u = f(\xi) = f(x - ct)$$

1.3 Lecture 3 - Conservation Laws

Recap:

- Solve PDEs of the form $a(x, y)u_x + b(x, y)u_y = 0$
- The char curves $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = a(x, y) \quad \frac{dy}{d\eta} = b(x, y)$$

- u is constant along chars.

ex Given $\frac{1}{x^2}u_x + u_y = 0$

$$\frac{dx}{d\eta} = \frac{1}{x^2} \quad \frac{dy}{d\eta} = 1$$

Method 1: divide to get parametric eq.

$$\begin{aligned} \frac{dy}{dx} &= x^2 \\ y &= \frac{1}{3}x^3 + \xi \end{aligned}$$

where constant ξ labels the char.

The general solution is $u = f(\xi) = f(y - \frac{1}{3}x^3)$ for an arbitrary function f . This can be confirmed by plugging u back into the PDE.

Method 2: solve parametric equations.

$$\begin{aligned}\frac{dx}{d\eta} &= \frac{1}{x^2} & \frac{dy}{d\eta} &= 1 \\ \Rightarrow \int x^2 dx &= d\eta & y &= \eta + B \\ \Rightarrow \frac{1}{3}x^3 &= \eta + A & y &= \eta + B\end{aligned}$$

We can absorb the two constants into one ξ

$$\begin{aligned}\Rightarrow \frac{1}{3}x^3 &= \eta + \xi & y &= \eta \\ \Rightarrow y &= \frac{1}{3}x^3 - \xi\end{aligned}$$

We should check that the chars cover the whole plane, i.e. no chars have been 'missed', via sketching or finding

$$\begin{aligned}\eta &= \eta(x, y) \\ \xi &= \xi(x, y)\end{aligned}$$

Coord transformation is also useful

$$\begin{aligned}\frac{1}{3}x^3 &= \eta + \xi \\ y &= \eta\end{aligned} \Rightarrow \begin{aligned}x &= (3\eta + 3\xi)^{\frac{1}{3}} \\ y &= \eta\end{aligned}$$

This transform our PDE

$$\begin{aligned}\frac{1}{x^2}u_x + u_y &= 0 \Leftrightarrow u_\eta = 0 \\ \text{Gen soln: } u &= f(\xi) \\ &= f\left(\frac{1}{3}x^3 - y\right)\end{aligned}$$

with arbitrary function f

ex Advection

Consider function $u(x, t)$ describing a quantity u at position x and time t

$$u_t + cu_x = 0 \quad (\text{Advection})$$

with constant $c > 0$

Chars $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \quad \frac{dx}{d\eta} = c$$

Solve

$$t = \eta + A$$

$$x = c\eta + B$$

Absorb into one constant using $A = 0, B = \xi$

$$\Rightarrow t = \eta$$

$$x = c\eta + \xi$$

$$\Rightarrow x = ct + \xi$$

By inverting, $\eta = t, \xi = x - ct$.

Now we have an transformed PDE

$$u_\eta = u_t + cu_x = 0$$

General solution is

$$\begin{aligned} u(x, t) &= f(\xi) \\ &= f(x - ct) \end{aligned}$$

for an arbitrary f .

Suppose we had initial condition $u(x, 0) = \phi(x)$, then $u(x, 0) = f(x) = \phi(x) \Rightarrow$ solution is $u(x, t) = \phi(x - ct)$.

Remarks:

We can solve more complicated PDEs using chars, for example,

$$u_t + cu_x = f(u)$$

Transform to coords (ξ, η) with $t = \eta$, $x = c\eta + \xi$

$$u\eta = f(u)$$

which can be solved as an ODE

Conservation Laws. (Logan 1.2)

- Where do PDEs come from?
- For a conserved quantity:

$$\begin{array}{ccc} \text{rate of change} & \implies & \text{net flow into domain} \\ \text{in domain} & & + \text{creation in domain} \\ \text{e.g. Animal population in a fixed region} & & \end{array}$$

Quantitatively

- Let $u(x, t)$ be density of some quantity in 1d domain.
- The amount of quantity in section dx is $u(x, t)Adx$.
- Let $\phi(x, t)$ be the flux, which is the rate of quantity crossing section x at time t . Note that $\phi(x, t)dt$ is the amount of stuff, and so is $u(x, t)dx$.
- $A\phi(x, t)$ is amount of quantity passing x at time t
- Let $f(x, t)$ be the rate at which quantity is created.
- $f(x, t)Adx$ is amount of quantity created in width dx per unit time.

Remarks:

- $\phi(x, t) > 0 \implies$ flux is L to R
- $\phi(x, t) < 0 \implies$ flux is R to L
- $f(x, t) > 0$ is source at x (creation)
- $f(x, t) < 0$ is sink at x (destroyed)

Balance:

change in
quantity in = flux + creation
section $[a, b]$

$$\begin{aligned}\Rightarrow \frac{d}{dt} \int_a^b u(x, t) A dx &= A\phi(a, t) - A\phi(b, t) + \int_a^b f(x, t) A dx \\ \Rightarrow \int_a^b u_t A dx &= - \int_a^b \phi_x A dx + \int_a^b f A dx \\ \Rightarrow \int_a^b (u_t + \phi_x - f) dx &= 0\end{aligned}$$

This is true for any domain $[a, b]$

$u_t + \phi = f$ where
change in
quantity = net flux + net creation

Remarks:

Can get a PDE for u by

- Specifying $f(x, t)$
- Choosing a relationship between ϕ and u, u_x , etc. This is called constitutive law.
- E.g. $\phi = cu \rightarrow$ advection.

Diffusion:

- Conservation law with no sources

$$u_t + \phi_x = 0$$

- Suppose $u(x, t)$ is density of a gas of particles. Then
 - Particles move from high u to low u .

- Flux increase with density gradient.
- Fick's law:

$$\phi(x, t) = -Du_x$$

where $D > 0$ is diffusion constant.

- This gives the diffusion equation

$$u_t = Du_{xx}$$

which is the same as the heat equation.

Remark:

- Can combine diffusion with other effects.
e.g. Advection-diffusion eq.

$$\text{flux: } \phi = cu - Du_x$$

Balance:

$$u_t + \phi_x = 0$$

$$u_t + cu_x = Du_{xx}$$

1.4 Lecture 4 - Diffusion Equation

Recap:

- Conservation Laws (Logan 1.2)
- Density: $u(x, t)$
- Flux: $\phi(x, t)$ (movement of u)
- Sources: $f(x, t)$ (creation of u)
- Balance in domain $[a, b]$

$$\frac{d}{dt} \int_a^b Au(x, t) dx = A\phi(a, t) - A\phi(b, t) + \int_a^b Af(x, t) dx$$

$$\int_a^b u_t dx = - \int_a^b \phi_x dx + \int_a^b f dx$$

$$\int_a^b (u_t + \phi_x - f) dx = 0$$

This holds for any $[a, b]$

$$u_t + \phi_x = f$$

change in
quantity + flux = source

Diffusion equation.

Suppose $u(x, t)$ is density/concentration of substance. We have Fick's Law for flux:

$$\phi(x, t) = -Du_x, \quad D > 0$$

- Movement from high u to low u
- Steeper gradient gives higher flux

Conservation/balance Law:

$$\begin{aligned} u_t + \phi_x &= 0 \\ \Rightarrow u_t &= Du_{xx} \end{aligned} \quad (\text{Diffusion/heat eq})$$

Remarks:

- D is called diffusivity or thermal conductivity.
- Some physical settings have a non-constant diffusivity $D(x)$

$$\phi = -D(x)u_x$$

This yields the PDE

$$\begin{aligned} u_t &= (Du_x)_x \\ &= D'u_x + Du_{xx} \end{aligned}$$

where $D' = \frac{dD}{dx}$

- Can also find nonlinear diffusion equations where $D = D(u)$

$$u_t = (D(u)u_x)_x$$

- We can combine diffusion with other terms, e.g. diffusion-advection

$$u_t + cu_x = Du_{xx}$$

$$\phi(x, t) = -Du_x + cu$$

Dimensions/units:

Suppose $u(x, t)$ is temperature, then

$$[u] = \text{temp}$$

$$[u_t] = \frac{\text{temp}}{\text{time}}$$

$$[u_{xx}] = \frac{\text{temp}}{\text{length}^2}$$

So $u_t = Du_{xx}$

$$\Rightarrow [D] = \frac{\text{length}^2}{\text{time}}$$

Consider heat flow in a region of length L . The time-scale T for temperature change is

$$T = \frac{L^2}{D} \text{ so } D = \frac{L^2}{T}$$

Boundary & Initial Conditions. Diffusion eqs usually have BCs and ICs. Consider heat flow in 1d with domain $0 < x < l$.

IC has form

$$u(x, 0) = u_0(x)$$

where $u_0(x)$ is initial temperature profile.

There are 3 types of common BCs.

1. Dirichlet BCs

Specify u at the boundary.

$$u(0, t) = g_1(t)$$

$$u(l, t) = g_2(t)$$

where g_1, g_2 are given i.e. u is known at boundary.

2. Neumann BCs

Specify u_x at the boundary.

$$u_x(0, t) = h_1(t)$$

$$u_x(l, t) = h_2(t)$$

where h_1, h_2 are given, i.e. flux is known at boundary.

3. Robin BCs

Specify u, u_x given at boundary.

$$au(0, t) + bu_x(0, t) = \psi_1(t)$$

$$au(l, t) + bu_x(l, t) = \psi_2(t)$$

where ψ_1, ψ_2 are given. E.g. Newton's Law of Cooling states that heat flow $\propto$ temp difference between object and environment.

$$bu_x(0, t) = -au(0, t) + \psi_1(t)$$

flux = object + environment

Remarks:

- A well-posed problem is a PDE with ICs and BCs s.t. it has a unique solution with a correct physical meaning (stability).
- The above terminology for BCs applies to general PDEs.

Steady-state solutions. Many time-dependent PDEs have solutions which approach steady states $u = u(x)$, which are time-independent. E.g. diffusion/heat equation:

$$u_t = Du_{xx}$$

Steady state solution $u(x)$ has $u_t = 0$, so

$$Du'' = 0 \quad (u' = u_x)$$

This gives a 2nd-order ODE, where

$$u(x) = Ax + B$$

for constants A, B set by BCs.

ex Consider the diffusion-decay equation

$$\begin{aligned} u_t &= Du_{xx} - ru \\ 0 < x < L, \quad t, D, r > 0 \end{aligned}$$

u has the flux $\phi = -Du_x$ and is destroyed at rate ru . Take BCs:

$$\begin{aligned} u(0, t) &= 0 \\ -Du_x(L, t) &= -1 \\ t &> 0 \end{aligned}$$

and IC

$$\begin{aligned} u(x, 0) &= g(x) \\ 0 < x < L \end{aligned}$$

Remarks:

This model could describe a diffusing pollutant (Du_{xx}) which decays at rate ru .

$$\begin{aligned} \text{At } 0 : u(0, t) &= 0 \\ &\Rightarrow u \text{ is held at } 0 \\ \text{At } L : -Du_x(L, t) &= -1 \\ &\Rightarrow \text{constant flux of } u \text{ into domain} \end{aligned}$$

Steady state problem:

$$u = u(x), \text{ so } u_t = 0$$

BVP:

$$\begin{aligned} Du'' - ru &= 0 \\ 0 < x < L \end{aligned}$$

BCs: $u(0) = 0, u'(L) = \frac{1}{D}$

Here we ignore IC and assume transients have decayed.

Solution of BVP:

$$\begin{aligned} u'' &= \frac{r}{D}u \\ u(0) &= 0, \quad u'(L) = \frac{1}{D} \end{aligned}$$

General solution:

$$u = A \sinh \left(x \sqrt{\frac{r}{D}} \right) + B \cosh \left(x \sqrt{\frac{r}{D}} \right)$$

consts A, B

BC1:

$$\begin{aligned} u(0) &= 0 \\ \Rightarrow B &= 0 \\ \Rightarrow u(x) &= A \sinh \left(x \sqrt{\frac{r}{D}} \right) \end{aligned}$$

BC2:

$$\begin{aligned} u'(L) &= \frac{1}{D} \\ \Rightarrow A \sqrt{\frac{r}{D}} \cosh \left(L \sqrt{\frac{r}{D}} \right) &= \frac{1}{D} \\ \Rightarrow u(x) &= \frac{\sinh \left(x \sqrt{\frac{r}{D}} \right)}{\sqrt{rD} \cosh \left(L \sqrt{\frac{r}{D}} \right)} \end{aligned}$$

Remarks:

- An ODE BVP may not have a solution. Physically the PDE may not have a steady state solution.
- The steady state solution could be unstable in general. (Here it turns out that $\lim_{t \rightarrow \infty} u(x, t) \rightarrow \text{steady state}$.)

Summary. Diffusion in 1d, for $u(x, t)$

$$\begin{aligned} u_t &= D u_{xx} \\ D &> 0 \end{aligned}$$

Can consider higher dimensions for $u(x, y, t)$

Preview: Vibrations and waves

Consider the 1d wave equation for $u(x, t)$:

$$u_{tt} = C^2 u_{xx}$$

This describes, for example, waves on a string. C has dimensions length/time and is the speed of waves. 2nd-order in time $\rightarrow$ waves (contrasts with 1st-order in time $\rightarrow$ diffusion).

We have the string fixed at length $x = 0, L$ and $u(x, t)$ is the height of the string at time t . And we have the following equation of motion for the string

$$u_{tt} = \sqrt{\frac{T}{\rho}} u_{xx}$$

where T is string tension, and ρ is the density of string.

1.5 Lecture 5 - Wave Equation

Recap:

- Diffusion: $u_t = D u_{xx}$, $D > 0$
- Wave: $u_{tt} = c^2 u_{xx}$, c is speed

Vibrations of a string.

- String fixed at $x = 0, x = l$.
- $u(x, t)$ is the height of the string at time t , position x
- Density $\rho = \frac{\text{mass}}{\text{length}}$

Assumptions:

- Small deflections, i.e. $|u_x|$ is small.
- Only force acting on string is tension (no gravity, damping, etc.)
- No horizontal motion of string (only vertical)

Let $T(x, t)$ be the tension, T acts tangent to the string.

Let $\theta(x, t)$ be the angle of the tangent to the string, so

$$\cos \theta = \frac{1}{\sqrt{1 + u_x^2}} \qquad \sin \theta = \frac{u_x}{\sqrt{1 + u_x^2}}$$

So T is the tension force along the string. Therefore, total force on the string in region $[a, b]$ is

$$\text{horiz: } T \cos \theta \Big|_a^b = \frac{T}{\sqrt{1 + u_x^2}} \Big|_a^b$$

$$\text{vert: } T \sin \theta \Big|_a^b = \frac{T u_x}{\sqrt{1 + u_x^2}} \Big|_a^b$$

We can simplify this using the small deflection assumption, $|u_x| \ll 1$

$$\begin{aligned} \text{Taylor series : } (1 + u_x^2)^{\frac{1}{2}} &= 1 + \frac{1}{2}u_x^2 - \frac{1}{8}u_x^4 + \dots \\ &\approx 1 \end{aligned}$$

where each u_x term gets smaller.

So the forces become

$$\begin{aligned} \text{horiz : } T \cos \theta \Big|_a^b &\approx T \Big|_a^b = \int_a^b T_x dx \\ \text{vert : } T \sin \theta \Big|_a^b &\approx T u_x \Big|_a^b = \int_a^b (T u_x)_x dx \end{aligned}$$

The accelerations are

$$\begin{aligned} \text{horiz : } &0 \quad (\text{by assumption}) \\ \text{vert : } &\int_a^b u_{tt} dx \end{aligned}$$

Using Newton's 2nd Law, $F = ma$

$$\begin{aligned} \text{horiz : } &\int_a^b T_x dx = 0 \\ \text{vert : } &\int_a^b (T u_x)_x dx = \int_a^b \rho u_{tt} dx \end{aligned}$$

This holds for all intervals $[a, b]$
 $\Rightarrow T_x = 0$ so $T(x, t) = T(t)$

and $(Tu_x)_x = Tu_{xx}$

$$\text{vert} : Tu_{xx} = \rho u_{tt}$$

$$\Rightarrow u_{tt} = c^2 u_{xx} \text{ for } c = \sqrt{\frac{T}{\rho}}$$

Remarks:

- Typically assume $T(t)$ is also time-independent so $T = \text{const}$, and speed $c = \sqrt{\frac{T}{\rho}}$ is const.
- For constant c , the general solution is

$$u(x, t) = F(x - ct) + G(x + ct)$$

where F is a right-moving wave, and G is a left-moving wave

- For a plucked string, our BCs are

$$u(0, t) = u(l, t) = 0$$

- To get a specific solution, we need 2 ICs, since the equation is 2nd order in time. E.g. we need

$$\text{Init position} : u(x, 0) = f(x)$$

$$\text{Init velocity} : u_t(x, 0) = g(x)$$

$$\text{for } 0 < x < l$$

- There are many variations, e.g.:

– Air resistance term, ru_t :

$$u_{tt} - c^2 u_{xx} + ru_t = 0$$

– External force, $f(x, t)$:

$$u_{tt} - c^2 u_{xx} = f(x, t)$$

Standing waves. Consider a vibrating string

$$u_{tt} = c^2 u_{xx}$$

$$\text{BCs : } u(0, t) = u(l, t) = 0$$

Standing waves are solutions of the form

$$u(x, t) = \phi(x) \cos \omega t$$

$$\text{or } \phi(x) \sin \omega t$$

$$\text{or in general } \phi(x) e^{i\omega t}$$

Each part of the string oscillates with:

- Same freq ω , in phase.
- Different amplitude $\phi(x)$

Solutions only exist for certain ω

$$\text{substitute } u = \phi(x) \cos \omega t \text{ into wave eq}$$

$$-\omega^2 \phi \cos \omega t = c^2 \phi'' \cos \omega t$$

$$\text{with } \phi(0) \cos \omega t = \phi(l) \cos \omega t = 0 \quad \forall t$$

This gives us an ODE BVP:

$$\phi'' = -\frac{\omega^2}{c^2} \phi, \quad \phi(0) = \phi(l) = 0$$

We have the following general solution

$$\phi(x) = A \cos \frac{\omega x}{c} + B \sin \frac{\omega x}{c}$$

$$\text{BC1: } \phi(0) = A = 0$$

$$\phi(x) = B \sin \frac{\omega x}{c}$$

$$\text{BC2: } \phi(l) = B \sin \frac{\omega l}{c}$$

$$B = 0 \text{ or } \sin \frac{\omega l}{c} = 0$$

If $B = 0$ then $\phi(x) = 0$ and $u(x, t) = 0$. We are interested in nonzero solutions so we need $\sin \frac{\omega l}{c} = 0$

Note: l and c are fixed by the problem, ω is the frequency of the standing wave. The condition $\sin \frac{\omega l}{c} = 0$ means the allowed frequencies ω of a standing wave satisfy $\sin \frac{\omega l}{c} = 0$

$$\Rightarrow \frac{\omega l}{c} = n\pi \text{ for } n \in \mathbb{Z}$$

Allowed frequencies: $\omega_n = \frac{n\pi c}{l}$

Modes: $\phi_n(x) = \sin \omega_n x = \sin \left(\frac{n\pi x}{l} \right)$, $n \in \mathbb{Z}$

- $B\phi_n(x)$ satisfies the BVP for any constant B , modes defined up to a multiplicative constant.
- $n = 0$ gives $\phi_n = 0$ so it is trivial.
- $n < 0$ gives the same mode as $n > 0$ up to a sign: $\phi_n(x) = -\phi_{-n}(x)$
- We can take the modes to have $n > 0$

$$\phi_n(x) = \sin \left(\frac{n\pi x}{l} \right), \quad n = 1, 2, 3, \dots$$

The standing waves are

$$\begin{aligned} u_n(x, t) &= \cos \left(\frac{n\pi ct}{l} \right) \sin \left(\frac{n\pi x}{l} \right) \\ &= \cos(\omega_n t) \phi_n(x), \quad n = 1, 2, 3, \dots \end{aligned}$$

Remarks:

- The BVP problem is an eigenvalue problem. The frequencies ω_n are the c -values and modes ϕ_n are eigenfunctions.
- Eigenfunctions ϕ_n are defined up to a constant multiplier, like eigenvectors.
- The frequencies ω_n are the fundamental frequencies of the string.

Quantum Mechanics (Logan 1.6)

Consider a particle in 1d. Classically, the particle has a position $x \in \mathbb{R}$. In quantum mechanics, the state of the particle is a function $\Psi(x, t) \in \mathbb{C}$ called the wave function. The probability density of finding the particle at x is $|\Psi(x, t)|^2$. Ψ obeys the Schrödinger equation (SE):

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} \quad (\text{free particle})$$

where $\hbar$ is Planck's constant

m is particle mass

If the 'energy cost' of being at position x is $V(x)$, then the SE is

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} + V(x)\Psi$$

Remarks:

- The particle is typically more likely to be at places where $V(x)$ is lower.
- Probability is conserved

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = 1 \quad \forall t$$

- Looking for solutions of the form

$$\Psi(x, t) = e^{-iEt/\hbar} \cdot \phi(x)$$

gives the following time-independent SE:

$$E\phi = -\frac{\hbar^2}{2m}\phi'' + V(x)\phi$$

Where E has the interpretation of 'energy'

For a particle in a '1d box,' $0 < x < l$, we have

$$E\phi = -\frac{\hbar^2}{2m}\phi'', \quad 0 < x < l$$

with the following BCs

$$\phi(0) = \phi(l) = 0$$

$\Rightarrow$ standing wave solutions